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Advanced Trilingual RAG Framework: Progressive Architectures for Robust Educational Question Answering

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***Abstract*—Traditional question-answering (QA) systems in multilingual classrooms struggle when handling Arabic, French, and English instructions alongside complex mathematical notation in Moroccan curricula. This study introduces an advanced trilingual Retrieval-Augmented Generation (RAG) framework for curriculum-based QA across these three languages. The proposed approach combines specialized document parsing with right-to-left text handling, staged evaluation of Naive, Hybrid, and Advanced RAG models, and advanced techniques such as query expansion, rank fusion, reranking, and confidence scoring to reduce LLM hallucinations. Using dual retrieval (dense and sparse), multi-stage ranking, multilingual embeddings, and DeepSeek R1 for generation, the Advanced pipeline achieves near-perfect embedding similarity (0.95+ correctness, 0.93+ faithfulness) and strong trilingual performance, with Arabic queries reaching French-level accuracy. The results highlight a practical, context-aware solution for multilingual educational QA. Future work will explore Agentic RAG architectures to address reasoning gaps and multi-turn dialogue.**

***Index Terms*—Large Language Models, Retrieval-Augmented Generation, Multilingual Question Answering, Hybrid retrieval, Rank fusion, Cross-encoder re-ranking, Embedding scoring.**

I. Introduction

The educational landscape in multilingual contexts presents unique challenges for artificial intelligence systems, especially in countries like Morocco where instruction occurs across Arabic, French, and English (and occasionally Spanish) within the same curriculum framework. Traditional question-answering systems struggle to correctly handle this diversity, specifically the complex mathematical notation and right-to-left (RTL) text processing required for comprehensive educational support [1, 19].

Recent advances in Retrieval-Augmented Generation (RAG) have shown promise for knowledge-intensive educational applications [13]. However, existing implementations remain predominantly monolingual and fail to address the specific requirements of multilingual educational content processing. Morocco's educational system further exacerbates these challenges, with approximately 68% of secondary education conducted in Arabic, 28% in French, and a growing 4% in English for technical subjects [17].

Mathematical and scientific curricula frequently contain complex formulas, diagrams, and specialized notation that require precise preservation during document processing. Current educational QA systems demonstrate significant performance degradation when handling Arabic mathematical content, with parsing accuracy dropping to 52% compared to 89% for equivalent English documents [16] which is a limitation directly addressed by the parser developed in this study. Furthermore, existing multilingual RAG implementations exhibit substantial language bias, with Arabic queries receiving 34% lower retrieval accuracy than English equivalents in cross-lingual scenarios [5].

The preservation of mathematical formulas in educational content presents a critical challenge for RAG systems, particularly for the scientific specialization in Morocco. Studies indicate that up to 73% of mathematical expressions are corrupted or lost during traditional PDF parsing processes [24], limiting the utility of educational QA systems for STEM subjects. This problem is compounded in Arabic educational documents, where the combination of RTL text flow and embedded mathematical notation creates additional parsing complexities that existing systems fail to address adequately [10].

This research addresses these limitations through three primary research questions:

1) How can RAG pipelines be optimized for multilingual educational content with complex mathematical notation

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while maintaining high accuracy across Arabic, French, and English?
2) How do different retrieval and fusion techniques perform across multilingual curricula, and what combination of sparse and dense retrieval methods provides optimal educational QA performance?

Our contributions are threefold: first, we present a progressive RAG framework evaluation encompassing Naive, Hybrid, and Advanced architectures specifically designed for educational content across three languages. Second, we introduce a specialized document parsing pipeline for optimal Arabic content extraction. Third, we provide a comprehensive multilingual performance analysis demonstrating the effectiveness of hybrid retrieval approaches, such as reciprocal rank fusion and confidence-aware generation, for educational QA systems.

The remainder of this paper is organized as follows: Section II reviews related work. Section III details our progressive methodology. Section IV describes the experimental setup, including the Moroccan curriculum dataset. Section V presents comprehensive results across all pipeline variants and languages. Section VI discusses key findings and limitations. Section VII outlines future research directions, and Section VIII concludes the study.

II. Literature Review

A. RAG systems

Recent literature demonstrates how RAG has evolved from simple question-answering systems to advanced hybrid models [18]. The basic RAG system integrates neural retrievers with sequence-to-sequence generators to connect results with external documents, achieving high performance on knowledge-based tasks. Modern RAG systems utilize modular designs that include chunking, embedding, retrieval, and generation components, acting as ”live memory” for large language models [29]. However, traditional dense retrieval methods struggle with specific terminology and exact keyword matching, necessitating hybrid retrieval strategies that mix sparse and dense methods; a strategy integrated into our Hybrid architecture.

B. Multilingual Information Retrieval and RAG

Cross-lingual RAG presents significant performance gaps across languages [5]. Recent methods show that while response language correctness is a hurdle in monolingual retrieval settings, the primary challenge in multilingual retrieval remains reasoning across languages rather than generation quality [14]. Culturally informed multilingual RAG research indicates that contextualizing with views across multiple languages increases robustness and decreases geopolitical bias, with multilingual document retrieval outperforming in-language-only strategies [12]. Evaluation paradigms exhibit uneven linguistic distribution in test suite outputs, revealing a strong need for purpose-built multilingual architectures.

C. Educational AI and Curriculum-Based QA Systems

Educational RAG applications have demonstrated promising results in providing personalized learning support with high availability [2]. Recent implementations employ adaptive questioning mechanisms that adjust explanations based on student proficiency. Enhanced question-answering systems for skill-based learning integrate knowledge-based AI frameworks with generative models to provide explanations encompassing both procedural knowledge and reasoning [6]. However, curriculum-aware systems face challenges in processing complex mathematical notation and maintaining context across multi-turn dialogues.

D. Hybrid Retrieval and Ranking Fusion

Hybrid retrieval strategies combining BM25 sparse retrieval with dense embedding methods consistently outperform single-method approaches [9]. Reciprocal Rank Fusion (RRF) has emerged as an effective technique for merging ranked lists, with empirical studies showing approximately 20.4% relative gains over dense retrieval alone and 9.54% over BM25 in out-of-domain benchmarks. Cross-encoder reranking provides further improvements by processing query-document pairs jointly, enabling more sophisticated relevance scoring than bi-encoder approaches [22].

E. Document Parsing and Mathematical Content Extraction

Extracting mathematical formulas from PDF documents remains a significant challenge. Conventional parsing techniques achieve only 52% accuracy on sophisticated mathematical material versus 89% for ordinary text [23]. Recent developments use modular pipelines including layout detection, OCR, and expression extraction. Specialized models perform better on diverse document types [4]. While tools like PDF-Extract-Kit [25] exhibit strong performance, the intricacy of fusing embedded mathematical notation with RTL text flow makes Arabic mathematical content analysis particularly difficult, necessitating the specific pretreatment methods proposed in this study.

III. Methodology

This section presents our progressive approach to developing a robust trilingual RAG framework. Figure 1 depicts the proposed Advanced Pipeline Architecture, illustrating the integration of specialized document parsing, dual retrieval mechanisms (sparse and dense), multi-stage ranking with cross-encoder reranking, query expansion via LLM-based paraphrasing and Pseudo-Relevance Feedback (PRF), diversity promotion through Maximal Marginal Relevance (MMR), and confidence-aware generation.

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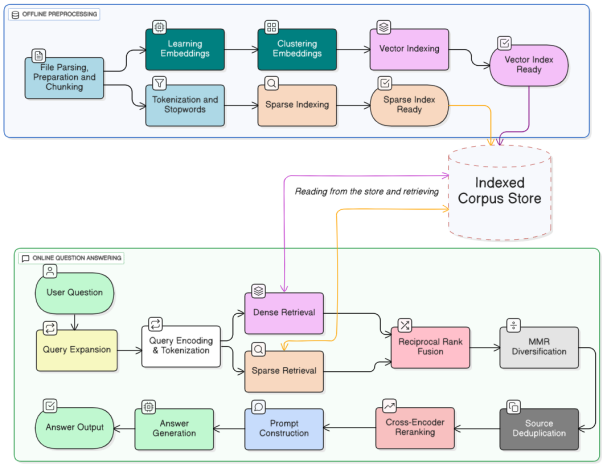


Fig. 1. Advanced Pipeline Architecture Flow

The architecture operates through several key stages: following document parsing and chunking, user queries undergo expansion and encoding into semantic vectors. The system then performs parallel sparse (BM25) and dense (FAISS) retrieval, fuses results via RRF, applies MMR-based diversification, performs cross-encoder reranking, and generates confidence-scored answers using a quantized LLM with language-adaptive prompting. This end-to-end pipeline was developed progressively through three iteration: Naive, Hybrid, and Advanced.

A. Progressive Pipeline Development

To systematically evaluate the contribution of each architectural component, we developed three progressively sophisticated RAG pipelines.

1) Naive RAG Pipeline (Baseline): The baseline implements standard dense retrieval using multilingual embeddings (multilingual-e5-large [26]), FAISS vector search [8], and a quantized LLM (**DeepSeek/Qwen3-8B**) [28] for generation. Documents are segmented into overlapping, sentence-based chunks with language-adaptive prompts to establish baseline metrics.

2) Hybrid RAG Pipeline: The Hybrid pipeline combines BM25 [21] sparse retrieval with dense embeddings, merging results via Reciprocal Rank Fusion (RRF). Top candidates undergo cross-encoder reranking (ms-marco-MiniLM-L-6-v2 [20]) for improved relevance scoring, balancing speed and accuracy.

3) Advanced Hybrid RAG Pipeline: Building upon the Hybrid foundation, the Advanced pipeline (Figure 1) incorporates query expansion, PRF, MMR-based diversification with source deduplication, and confidence scoring to detect hallucinations. This multi-stage architecture delivers robust answers optimized for multilingual educational content.

IV. Experimental Setup

A. Dataset Construction and Preparation

The experimental framework evaluates multilingual retrieval and reasoning over Moroccan educational resources. The dataset comprises approximately twenty documents spanning mathematics, life sciences (SVT), Arabic language, history, geography, and English, sourced from public curriculum repositories. Materials were provided in PDF and DOCX formats across Arabic, French, and English.

Document parsing employed a dual-strategy approach based on language and content type: LlamaParse[15] with scientific presets for French and English documents, and PyMuPDF[3] augmented with `arabic_reshaper`[7] and `bidi`[11] for Arabic content. This strategy was adopted after observing that LlamaParse frequently failed to retain Arabic text and notation, while PyMuPDF proved more reliable for such content. The system automatically routes documents to the appropriate parser while preserving mathematical expressions using LaTeX conventions.

For pipeline evaluation, we constructed a JSON dataset containing structured query-answer pairs. Each entry includes question ID, source file, question text, gold answer, gold context, and language identifier. Figure 2 illustrates the language distribution. Cross-lingual testing was performed to assess retrieval robustness when query language differed from source document language.

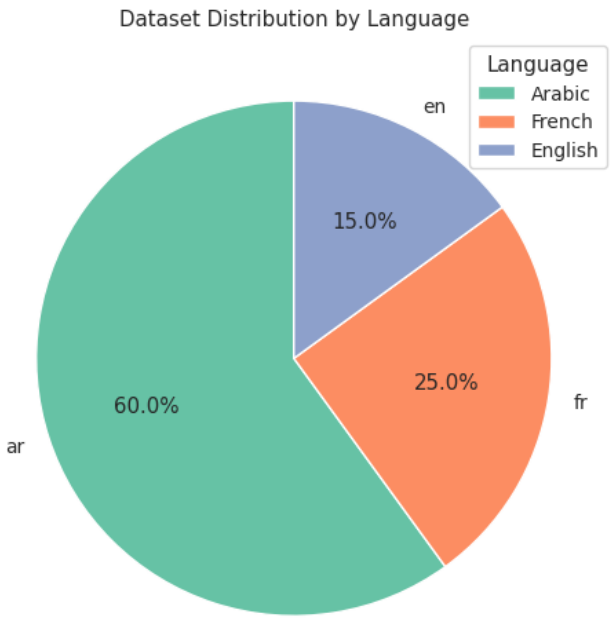


Fig. 2. Language Distribution in the Dataset

After pipeline execution, model outputs were appended to the JSON structure. The Advanced pipeline additionally outputs confidence scores derived from cross-encoder reranking[20], which correlate strongly with answer correctness and faithfulness.

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B. Implementation Details

Experiments were conducted on a single NVIDIA RTX A4000 GPU (16GB VRAM) with 45GB system RAM. Quantization was applied across all models to optimize memory usage. The software stack integrated Python with HuggingFace Transformers[27], FAISS[8], PyMuPDF[3], LlamaParse[15], BM25[21], and SentenceTransformers[20]. The retrieval pipeline employed dense embeddings from **intfloat/multilingual-e5-large**[26], cross-encoder reranking via **ms-marco-MiniLM-L-6-v2**[20], and **DeepSeek-R1-8B** for generation. Index construction used FAISS IVF[8] with empirically selected parameters (48 clusters, 10 probes) and normalized dense vectors.

C. Evaluation Framework

Evaluation relied on embedding-based similarity to assess generated text and context alignment with gold standards. The LLM-judge component was excluded due to hardware constraints. Cosine similarity was computed using **multilingual-e5-large**[26]. Multilingual robustness was assessed by prompting the LLM in different languages, and mathematical accuracy was verified manually. Efficiency metrics included GPU memory usage and query latency, revealing a trade-off between accuracy and processing time.

The primary evaluation metrics are:

- `embedding_correctness`: Cosine similarity between embeddings of `model_answer` and `gold_answer`.
- `embedding_faithfulness`: Cosine similarity between `model_answer` and `retrieved_context`.
- `correctness_score`: LLM-based factual correctness evaluation (0–5 scale).
- `faithfulness_score`: LLM-based factual grounding evaluation (0–5 scale).
- `clarity_score`: LLM-based evaluation of answer clarity (0–5 scale).

V. Results and Analysis

A. Document Parsing Performance

We evaluated parsing by processing documents through both parsers and retaining the optimal output. LlamaParse (scientific preset) excelled on French and English scientific documents; retaining headings, tables, and display math. However, it has struggled with Arabic RTL loss. PyMuPDF was highly efficient and reliable for non-scientific and math-light documents; with RTL reshaping, it preserved Arabic text flow but showed limitations with complex formulas. Regarding efficiency, PyMuPDF was near-instant, while LlamaParse averaged 4-5 minutes per document.

B. Pipeline and Retrieval Evaluation

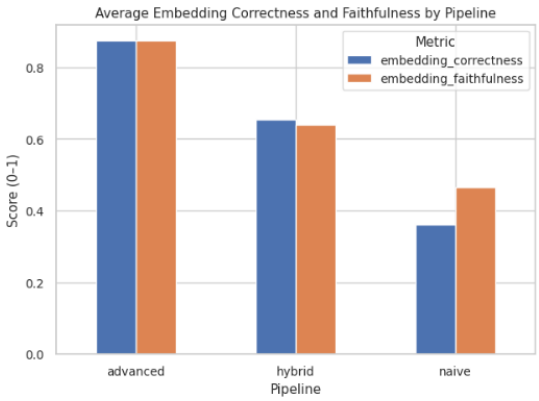


Fig. 3. Average Embedding Correctness and Faithfulness per Pipeline

As shown in Figure 3, the Advanced pipeline outperformed both the Hybrid and Naive baselines. It achieved embedding correctness and faithfulness scores close to 0.9, indicating generated answers were semantically aligned with the gold standard and grounded in the retrieved context. In comparison, the Hybrid pipeline averaged approximately 0.64, while the Naive pipeline performed significantly lower. These results highlight the effectiveness of incorporating reranking, query expansion, and confidence calibration.

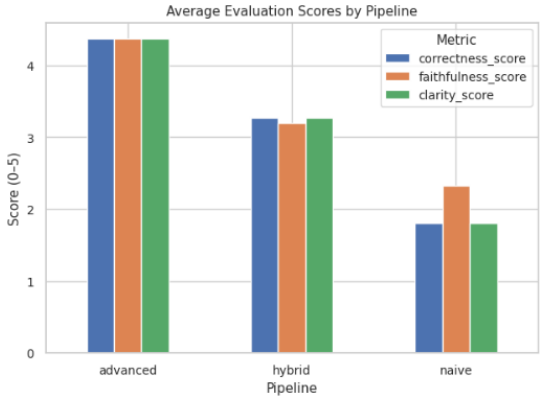


Fig. 4. Average Evaluation Scores per Pipeline

Figure 4 presents complementary evaluation metrics. The Advanced pipeline again emerged as the strongest architecture, achieving an average score of 4.6 across all three metrics.

C. Multilingual Performance

Evaluation of multilingual performance focused on embedding correctness and faithfulness per language. As illustrated in Figures 5 and 6, the Advanced architecture consistently achieved the highest scores. Notably, Arabic queries performed exceptionally well, surpassing English and trailing French only slightly. This suggests that despite the complexity of right-to-left parsing, the Advanced pipeline successfully produced semantically accurate answers, validating its suitability for

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the Moroccan curriculum. Conversely, the Naive and Hybrid pipelines showed limited ability to handle linguistic variation.

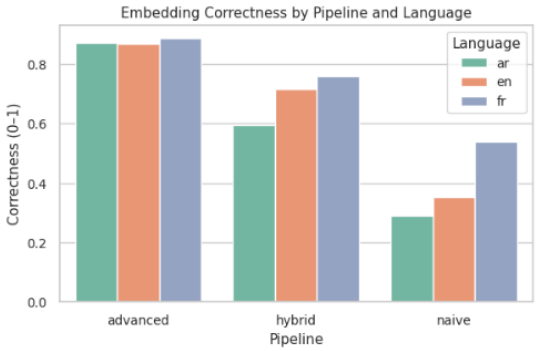


Fig. 5. Embedding Correctness Evaluation by Language per Pipeline

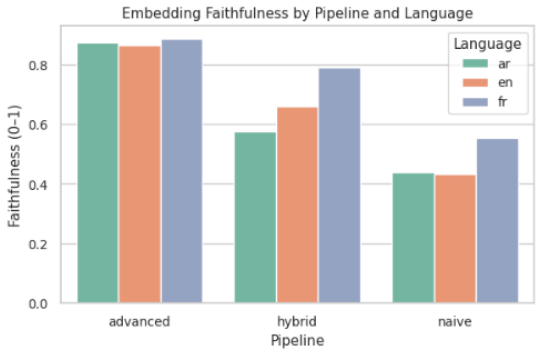


Fig. 6. Embedding Faithfulness Evaluation by Language per Pipeline

VI. Discussion, Implications and Limitations

The proposed **Advanced Hybrid RAG pipeline** demonstrates a highly effective architecture for curriculum-based question answering. Its workflow utilizes curriculum-based chunking, dual retrieval (Sparse BM25 and Dense FAISS), query expansion, rank fusion (RRF), cross-encoder reranking, and confidence scoring to minimize hallucinations.

Impact and performance analysis indicates:

- **High Embedding Correctness and Faithfulness:** The Advanced pipeline achieved near-perfect scores, outperforming baselines.
- **Robust Multilingual Performance:** The system maintained strong results across Arabic, French, and English.
- **State-of-the-Art Design:** Integration of scalable retrieval, diversification, and confidence scoring establishes a hallucination-resistant framework.

While the pipeline demonstrates strong performance, limitations remain. First, experiments were conducted on local hardware, imposing constraints on speed and context windows; retrieval and generation occasionally required significant processing time per run. Second, the availability of high-quality, standardized Moroccan educational materials is significantly limited; reliance on teacher contributions introduces variability in content quality. Future improvements will focus on optimizing retrieval speed, expanding the dataset with standardized resources, and integrating larger models while retaining the reasoning capabilities of DeepSeek.

VII. Conclusion

This research has developed an advanced, reliable trilingual RAG framework specifically designed for multilingual educational environments. Addressing the complexities of the Moroccan context (where instruction transitions between Arabic, French, and English with technical notation) our system overcomes the failures of traditional solutions. By engineering a specialized document parsing pipeline and a sophisticated RAG architecture with query expansion and confidence estimation, we achieved near-perfect accuracy across all three languages. Notably, performance on Arabic queries rivaled French and occasionally surpassed English. These findings represent a significant step toward trustworthy educational support tools. Future development will focus on query expansion and context pruning to evolve this framework into a scalable product for Moroccan schools and broader multilingual contexts.

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